Boxplot Overview

Vinny Paris

August 2026

I'm writing this as a general outline as that is how I normally organize my thoughts with respect to the course organization. Generally, pdf powerpoints will be posted in a more "classic" fashion.

1 Statistics as a field

Statistics is the study of variability

- We do NOT study deterministic systems like y = 2x + 3
- Instead we study things like dice rolls where the same act (rolling the dice) produce variable outputs
- Generally, a goal is to distinguish between variability due to random noise vs structural differences
  - Eg people's heights randomly vary (random noise) but the data and science suggest males are slightly taller than females (structural difference)

2 Population vs Samples

1. *Populations* are everything you are interested
   - Eg ALL CofC students
   - Gathering information on the entire population is called a *census*
     - Usually not plausible unless your population is small or your resources are immense
     - Eg The US census talks to everyone inside the USA once every ten years
     - Eg A census of all monarch butterflys would be impossible
2. A *parameter* is a numeric summary about the population
     - Eg average SAT score of all CofC students
     - Very hard to calculate because it's hard to collect a census
3. A *sample* is a subset/smaller portion of the entire population
   - Eg my class is a subset of all CofC students, that is it's a sample
   - Ideally, samples look like the population
     - We achieve this through random sampling
     - We will get to this later
4. A *statistic* is a numeric summary of the sample
   - Eg the average SAT score of my students
   - If the sample looks like the population, the statistic should look like the parameter
   - We use the statistic to estimate/say something meaningful about the (unknown) parameter

3 5 number summaries

See the example below the outline for this worked out to help make some this make more sense

1. A *percentile* is the value that BLANK% of the population is below
   - Eg the 82nd percentile for SAT math scores
2. The five number summary reports very specific percentiles of a given data set
   - minimum (0th percentile)
     - Smallest value in the data set
   - maximum (100th percentile)
     - Largest value in the data set
   - median (50th percentile)
     - The "middle" number
     - Half of the values should be above this value and half should be below this value
     - It can be found by...
       (a) Arrange the numbers in increasing order (if they aren't already)
       (b) Count how many numbers there are, call this n
       (c) Find the $\frac{n+1}{2}^{th}$ number, eg the 8th number in the line up
       (d) If it's a whole number the median is in your data set
       (e) If $\frac{n+1}{2}$ is a fraction (eg 8.5) then you take the average of the numbers $\frac{n+1}{2}$ is between (eg average the 8th and 9th number)
   - Q_1 (25th percentile)
     - This is the median of the lower half of the data
     - Divide your data in half at the median
       * If you median isn't in your data don't include it
       * If the median IS in your data set, DO include it in the lower half of the data (and the upper half for Q_3)
         · I know there are other ways to calculate this but I honestly don't care. There are many ways to calculate any value in statistics (eg I know 5 different ways to calculate the mean of a data set). Having seen a different technique before just means you are already hip deep in how statistics is not just math
         · See the Wikipedia page for quartiles for four different calculation methods
   - Q_3 (75th percentile)
     - Median of the upper half of the data
     - Similar to Q_1, if you median isn't in your data don't include it
     - If the median IS in your data set, DO include it in the upper half of the data (see Q_1 for commentary)

4 Example 1

$$1,\ 5,\ 6,\ 6,\ 7,\ 8,\ 9,\ 10$$

So the first two easy ones...

$$\begin{aligned} min &= 1 \\ max &= 10 \end{aligned}$$

We now need to find the median so...

1. There are 8 numbers in total
2. $n = 8$
3. $\frac{n+1}{2} = \frac{8+1}{2} = 4.5$
3. So we need the average of the 4th and 5th number

$$median = \frac{6+7}{2} = 6.5$$

Okay, now the hard two. Let's tackle Q_1 first..

1. First we divide the data in half and take the lower half
2. We divide at 6.5 so we need the median of $1,\ 5,\ 6,\ 6$
3. So $n = 4$ so $\frac{n+1}{2} = \frac{4+1}{2} = 2.5$
4. So its the average of the second and third number...

$$Q_1 = \frac{5+6}{2} = 5.5$$

Finally, for Q_3...

1. 7,8,9,10 is the upper half of the data
2. Finding the median of this ends up being the 2.5th number again so...

$$Q_3 = 8.5$$

5 Example 2

$$-3, -1, 0, 1, 2, 4, 5$$

$$\begin{aligned} min &= -3 \\ max &= 5 \\ median &= 1 \\ Q_1 &= -.5 \\ Q_3 &= 3 \end{aligned}$$

Note that Q_1 is the median of -3, -1, 0, AND 1. The median is included in both the lower and upper half if the median is in the data set. As such, Q_3 is the median of 1, 2, 4, and 5.

6 Boxplots

These are the graphical representation of the five number summary

- The left part of the box is Q_1
- The right part of the box is Q_3
- The median is a line in the box
- The left whisker is Q_1 - 1.5 * IQR pulled UP to the next data point above the whisker
  - IQR = Inter Quartile Range = Q_3 - Q_1
- The right whisker is Q_3 + 1.5 * IQR pulled DOWN to the next data point below the whisker
- Any data points that are lower than the lower whisker or higher than the upper whisker are plotted as single points (usually labeled)
- NEVER stretch the whisker below Q_1 - 1.5 * IQR nor above Q_3 + 1.5 * IQR
- NEVER let the whiskers end where there is not a data point. Otherwise it looks like the boxplots are always symmetric
- ALWAYS end a whisker on a data point
- Symmetry vs Skewed: IF the boxplot can be folded in on itself and roughly line up we say the data is *symmetric*. If one side is stretched out we called it skewed in that direction.
- The graph below is skewed right and has outliers